International Cybersecurity and Digital Forensics Academy (ICDFA)

Lab 1 - Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis

DIGITAL FORENSICS INVESTIGATION REPORT

Case: ICDFA Chapter 01 Investigation - Evidence Recovery & Analysis

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Course Code	SBTD – DF202
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Date:	31-08-2026
Environment:	Kali Linux
Case Reference:	ICDFA_Ch01_Investigation (Target: INCOME.XLS)

1. Executive Summary

A digital forensic acquisition and analysis was performed on the raw disk image Ch01InChap01.dd to investigate deleted files and evaluate evidence recovery methodologies across both Graphical User Interface (GUI) and Command Line Interface (CLI) forensic environments.

Through comprehensive analysis using Autopsy Forensic Browser (v2.24) and The Sleuth Kit (TSK) command-line suite, multiple deleted artifacts were identified on the raw FAT12 volume. Specifically, the target file Income.xls (Metadata Inode 13) was successfully identified, analyzed, and extracted using three independent recovery techniques: (1) Web-based GUI export via Autopsy, (2) Direct inode extraction via icat, and (3) Automated bulk partition recovery via tsk_recover. Cryptographic hashing (MD5 and SHA-256) confirmed 100% data integrity across all extracted instances, demonstrating complete non-repudiation and tool consistency.

2. Lab Objectives

- Verify the cryptographic integrity of the primary raw disk image using pre-analysis MD5 hash matching.
- Mount and inspect raw/unpartitioned volume images using web-based forensic suites (Autopsy).
- Identify, locate, and export deleted directory entries and orphaned file clusters.
- Perform low-level file system structure analysis (FAT12) using The Sleuth Kit (TSK) command-line interface.
- Extract specific target files via metadata address (icat) and validate recovery through cryptographic hash comparison (MD5 / SHA-256).

3. Tools & Environment

Tool / Software	Version / Details	Purpose
Kali Linux	Rolling / VM	Primary analysis operating system environment
Autopsy Forensic Browser	v2.24	Web-based digital forensics GUI platform
The Sleuth Kit (TSK)	v4.x CLI Utilities	Low-level file system analysis (fsstat, fls, istat, icat, tsk_recover)
GNU Coreutils	Standard Linux	Hash calculation (md5sum, sha256sum) and submission packaging (zip)

4. Evidence Details & Verification

Evidence File Name: Ch01InChap01.dd
File Size: 1,474,560 bytes (1.44 MB Floppy Disk Image)
File System Type: FAT12 (fat12)
Verified Baseline MD5 Hash: a117773bcf1fc88ec0ab8e0a349fbbcb

Fig 4.0 Autopsy interface verifying successful import of Ch01InChap01.dd as a Volume Image (FAT12) with MD5 hash validation

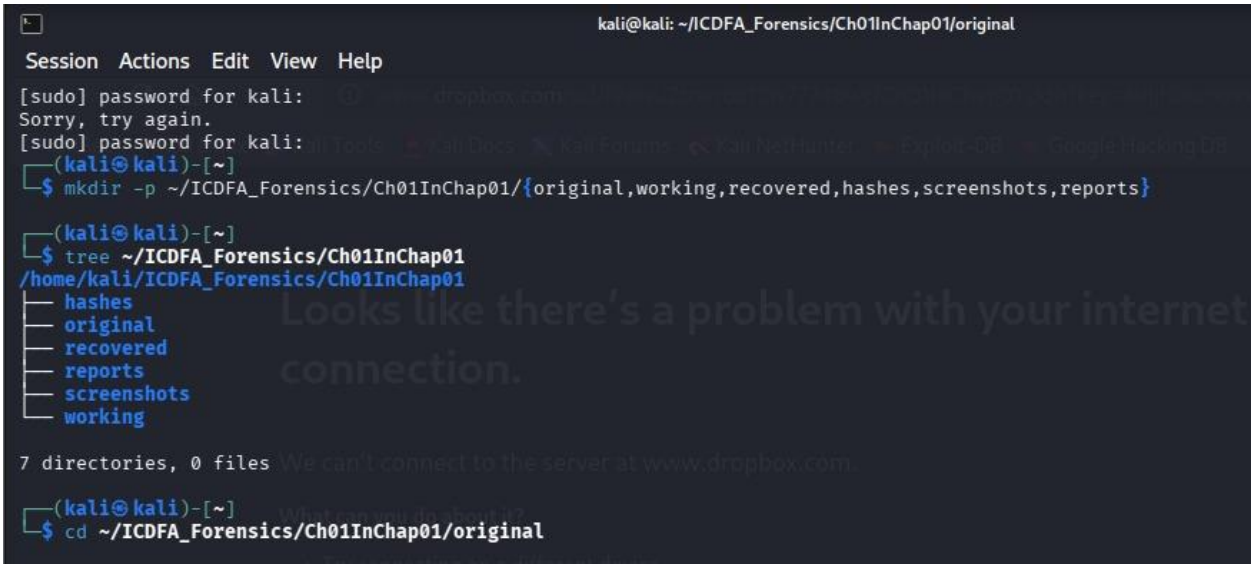


Fig 4.1 Initial workspace directory structure setup and verification for Case ICDFA Chapter 01 using mkdir and tree commands in Kali Linux.

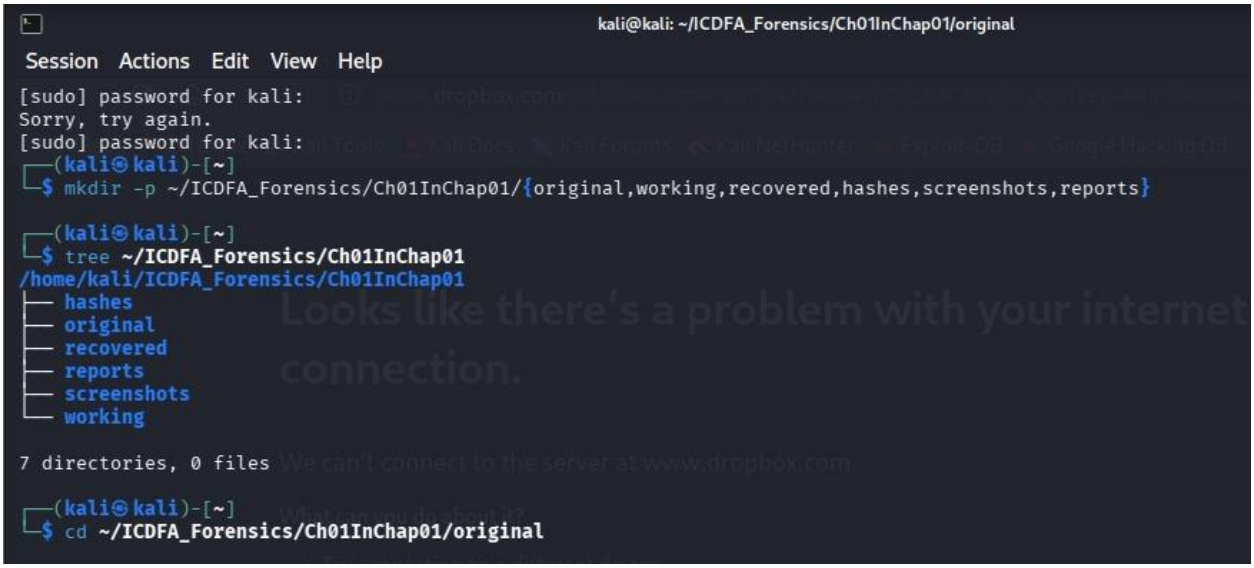


Fig 4.2 Evidence file verification confirming Ch01InChap01.dd is a 1.5MB raw floppy disk image parsed as a DOS/MBR boot sector with a FAT12 file system.

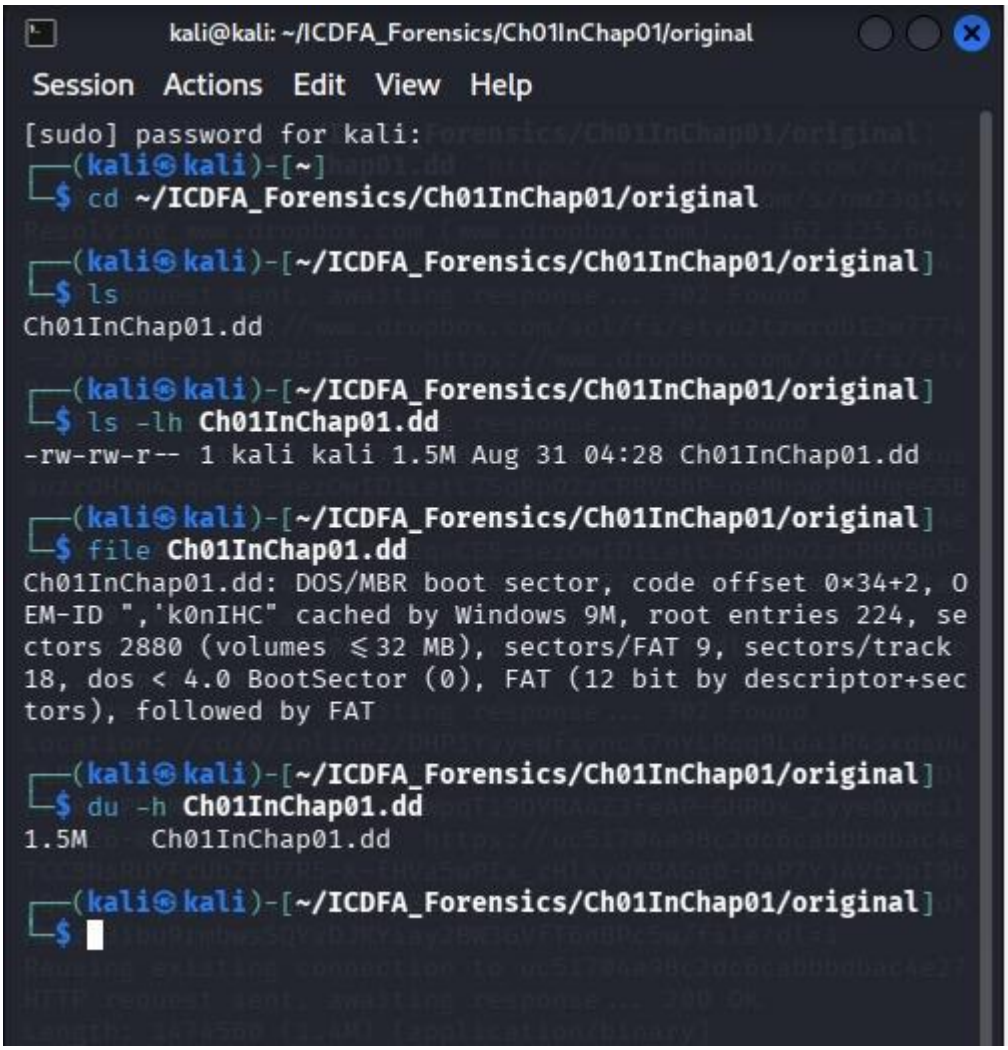
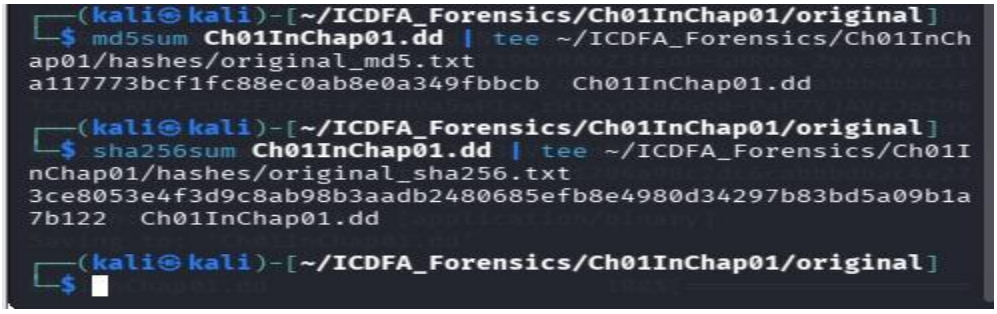


Fig 4.3 Cryptographic verification establishing the original evidence hash values (MD5 and SHA-256) and outputting logs to the `hashes` directory to ensure forensic integrity.



5. Technical Methodology & Forensic Analysis

Phase 1: GUI Analysis & Evidence Export (Autopsy)

1. Case & Host Setup: Created case ICDFA_Ch01_Investigation and host host1.
2. Volume Classification: Imported Ch01InChap01.dd as a Volume Image (raw partition) rather than a partitioned disk image. Autopsy successfully parsed the file system as FAT12.
3. Directory Browsing: Navigated to the File Analysis view and identified deleted entries flagged in red text (r/r):
   - confirmation.txt
   - Income.xls
   - letter1.txt
   - Regrets.doc
4. Target Extraction: Selected Income.xls and executed Export File, saving the file locally as autopsy_Income.xls (13,824 bytes).

Fig 5.1.0 Configuring the "ADD A NEW IMAGE" form in Autopsy web interface by entering the evidence path (`. . ./original/Ch01InChap01.dd`) for forensic analysis.

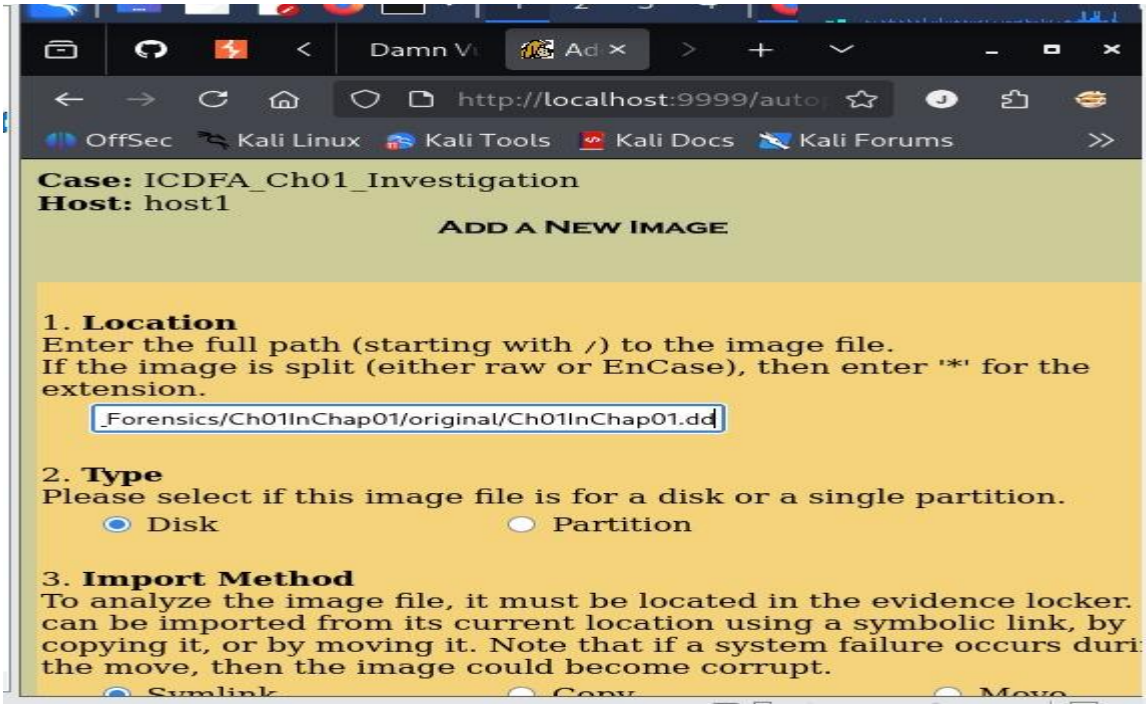


Fig 5.1.1 Configuring Autopsy image integrity verification by attaching the MD5 hash and confirming the partition layout as FAT12 (`fat12`).

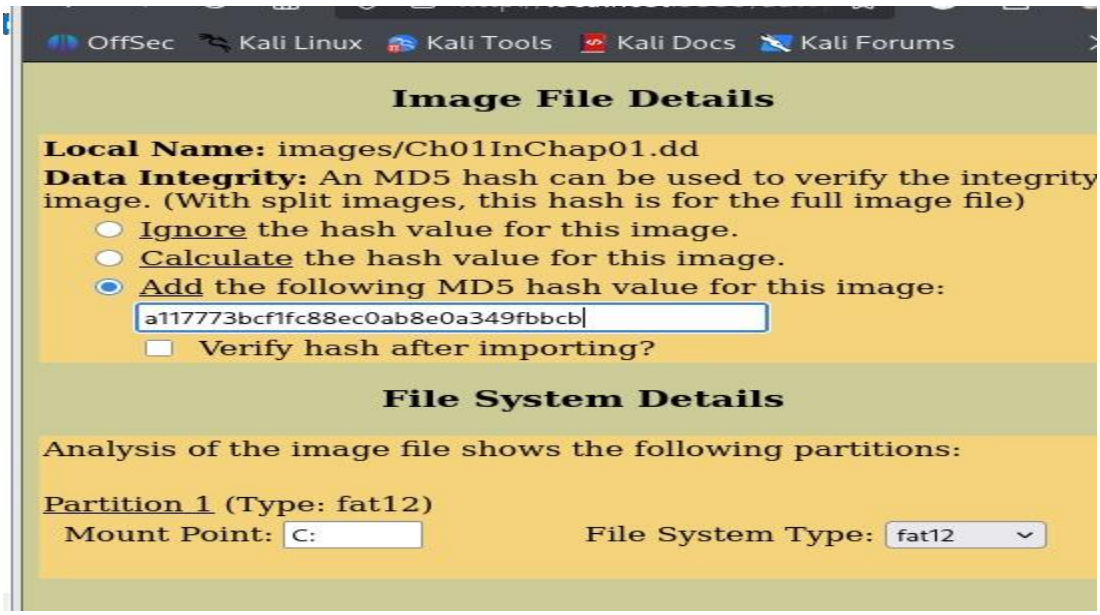


Fig 5.1.2 Autopsy File Browsing Mode interface displaying deleted files recovered from the FAT12 root directory `C:/`.

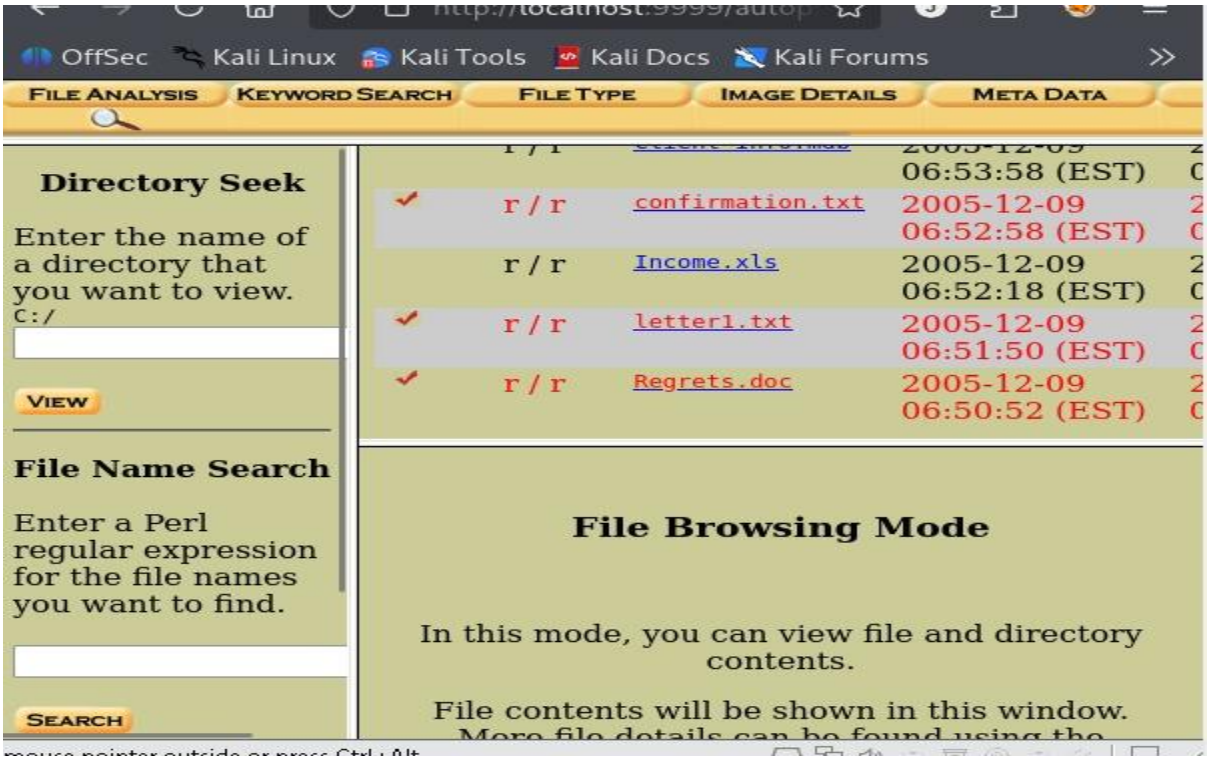
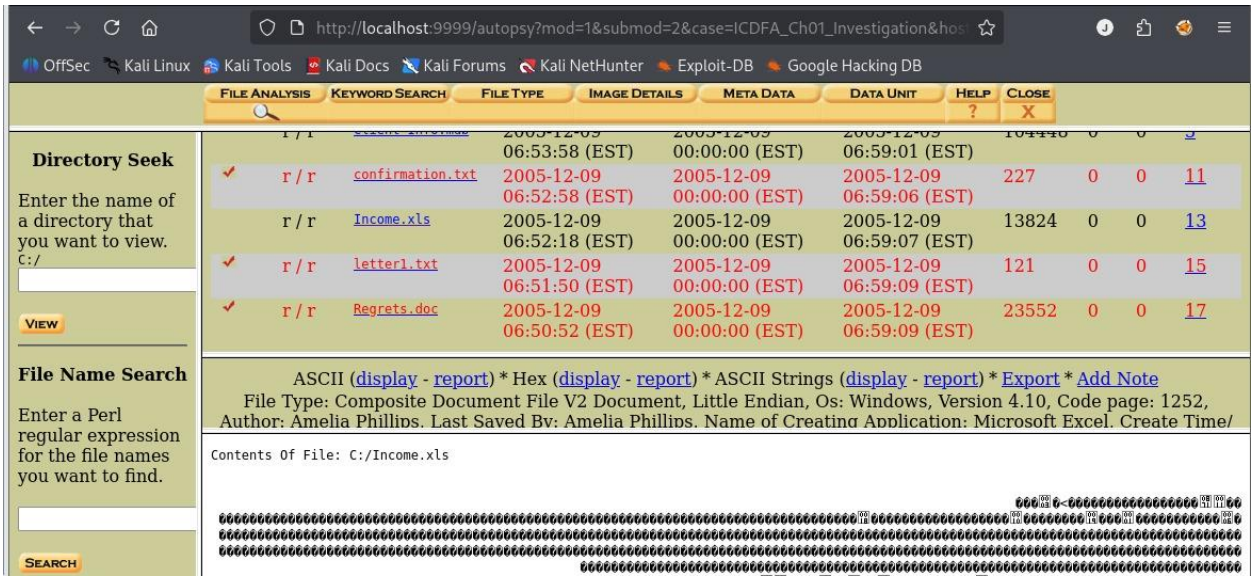


Fig 5.1.3 Autopsy File Analysis view listing deleted files on the FAT12 root directory, highlighting `Income.xls` (Inode 13, 13,824 bytes) and displaying document structural metadata.



Phase 2: Command Line Forensic Analysis (The Sleuth Kit)

Fig 5.2.0 Organizing exported GUI artifacts. Executing `mv` transfers the recovered spreadsheet file from the browser download directory to `recovered/autopsy_Income.xls`, followed by `ls recovered` verification to ensure clean evidence management.

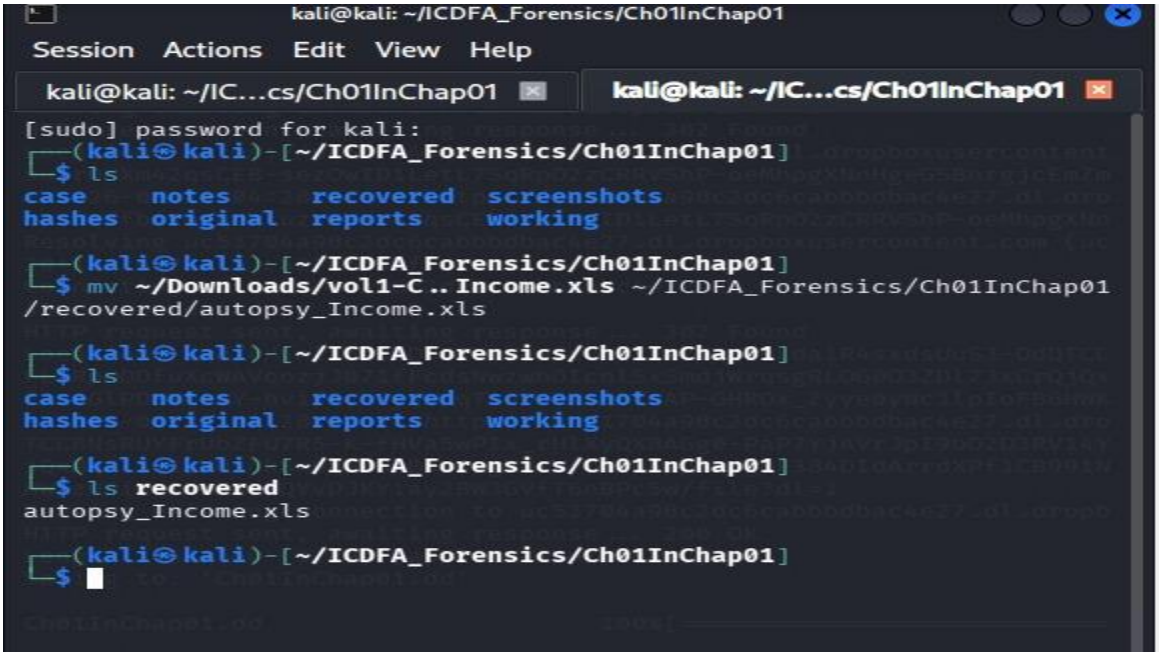


Fig 5.2.1 fsstat Analysis: Ran fsstat ../original/Ch01InChap01.dd to confirm FAT12 parameters, sector size (512 bytes), and cluster layout.

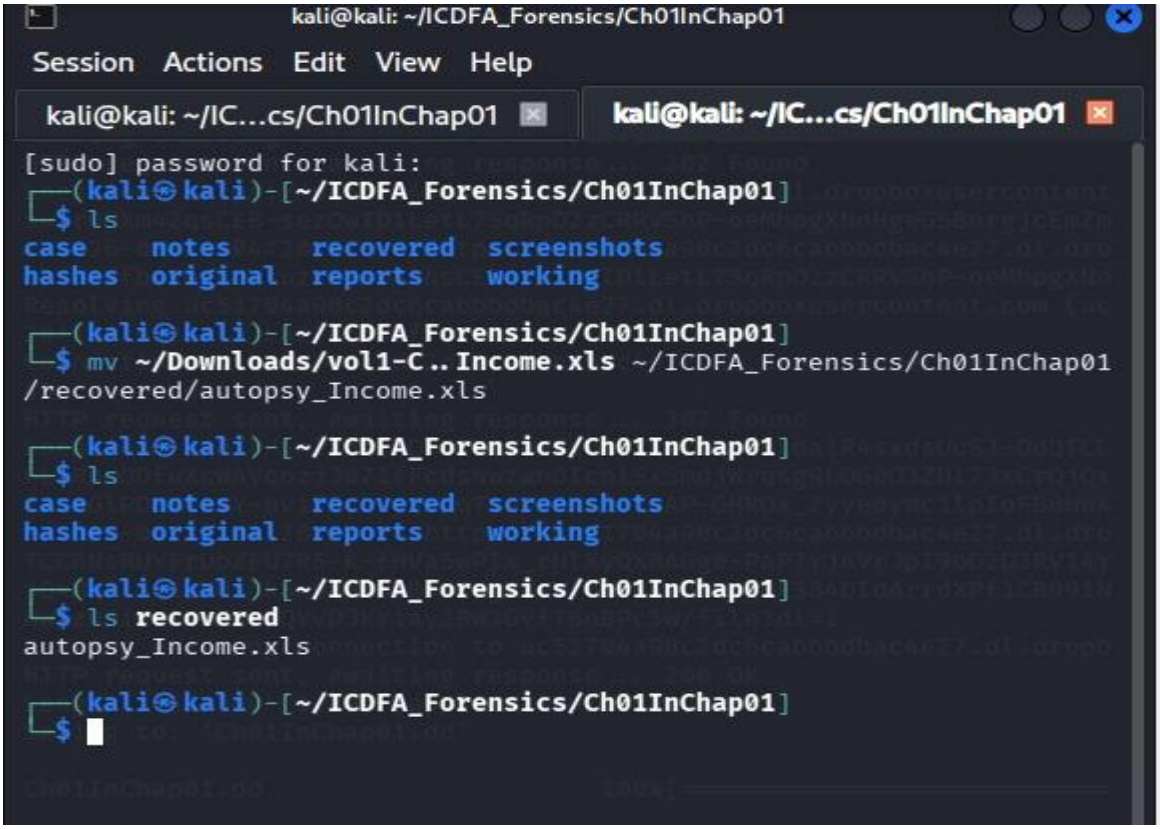


Fig 5.2.2 fls Analysis: Ran fls -r -d ../original/Ch01InChap01.dd to enumerate deleted records. Located Income.xls mapped to Metadata Inode 13.

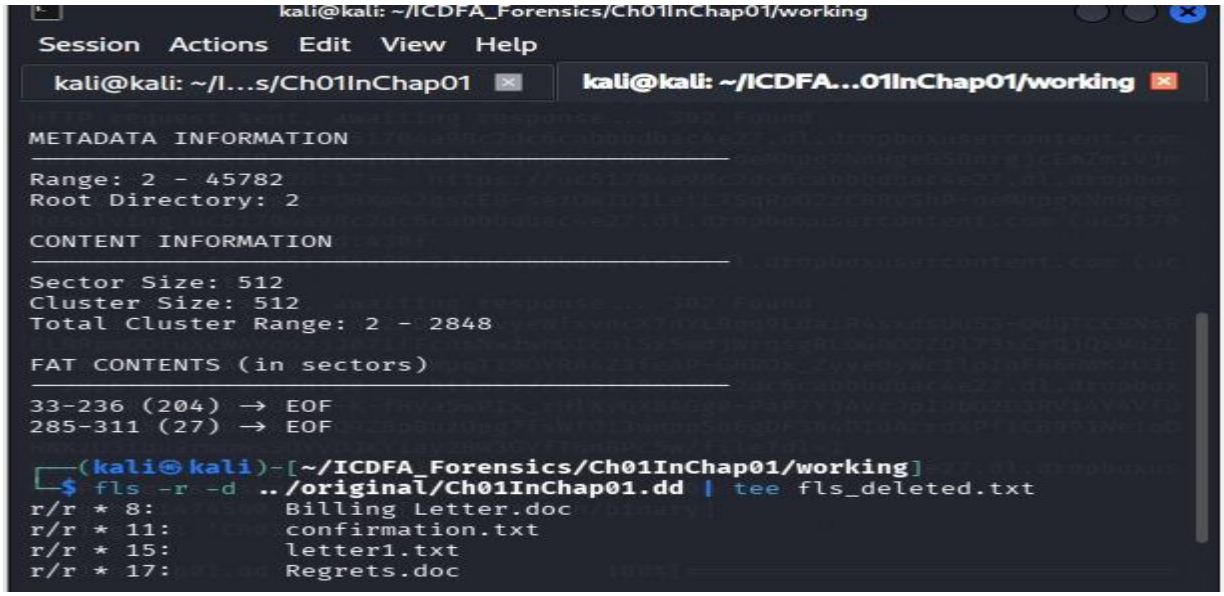


Fig 5.2.3 istat Analysis: Ran istat ../original/Ch01InChap01.dd 13 to inspect file attributes, record size (13,824 bytes), and sector allocation sequence.

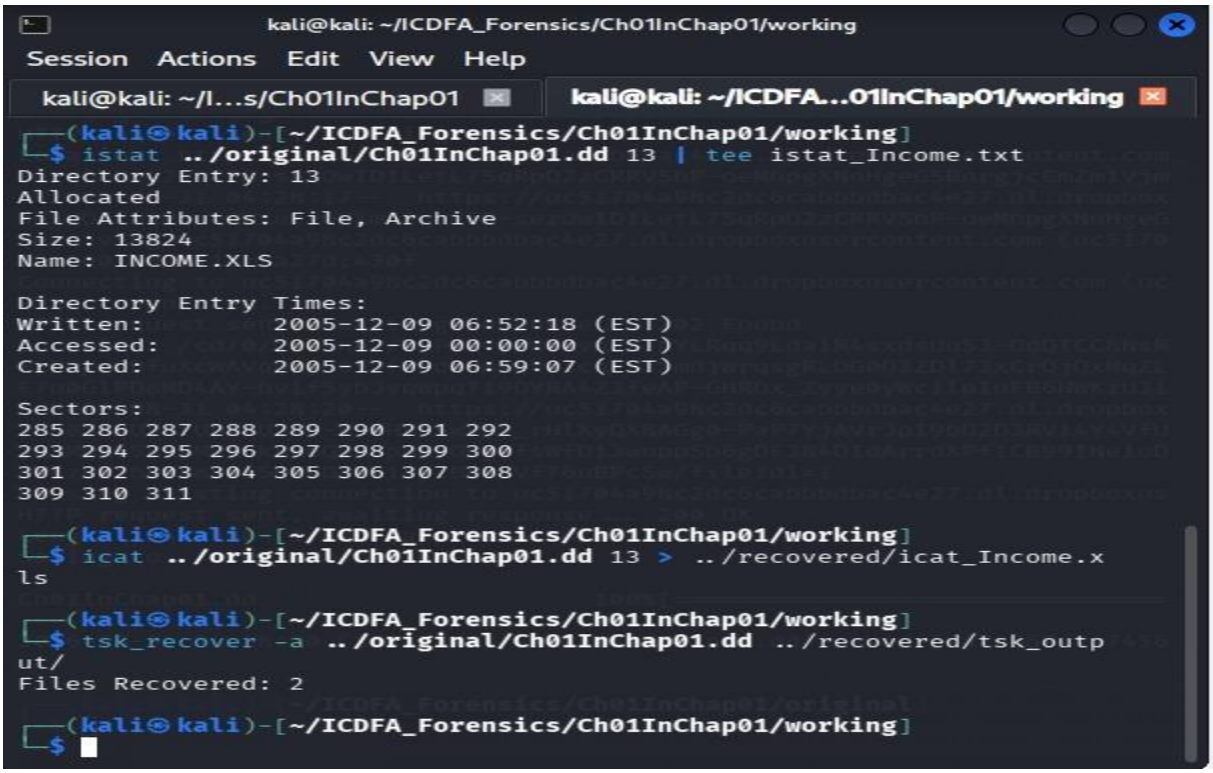


Fig 5.2.4 Direct Extraction (icat): Executed icat ../original/Ch01InChap01.dd 13 > ../recovered/icat_Income.xls.

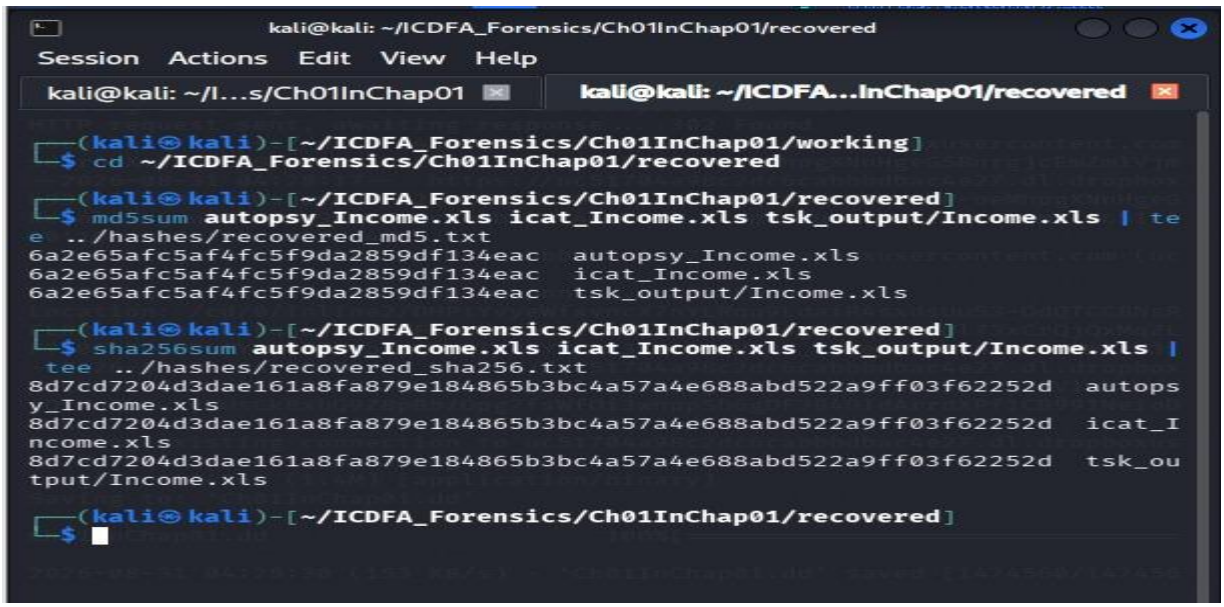
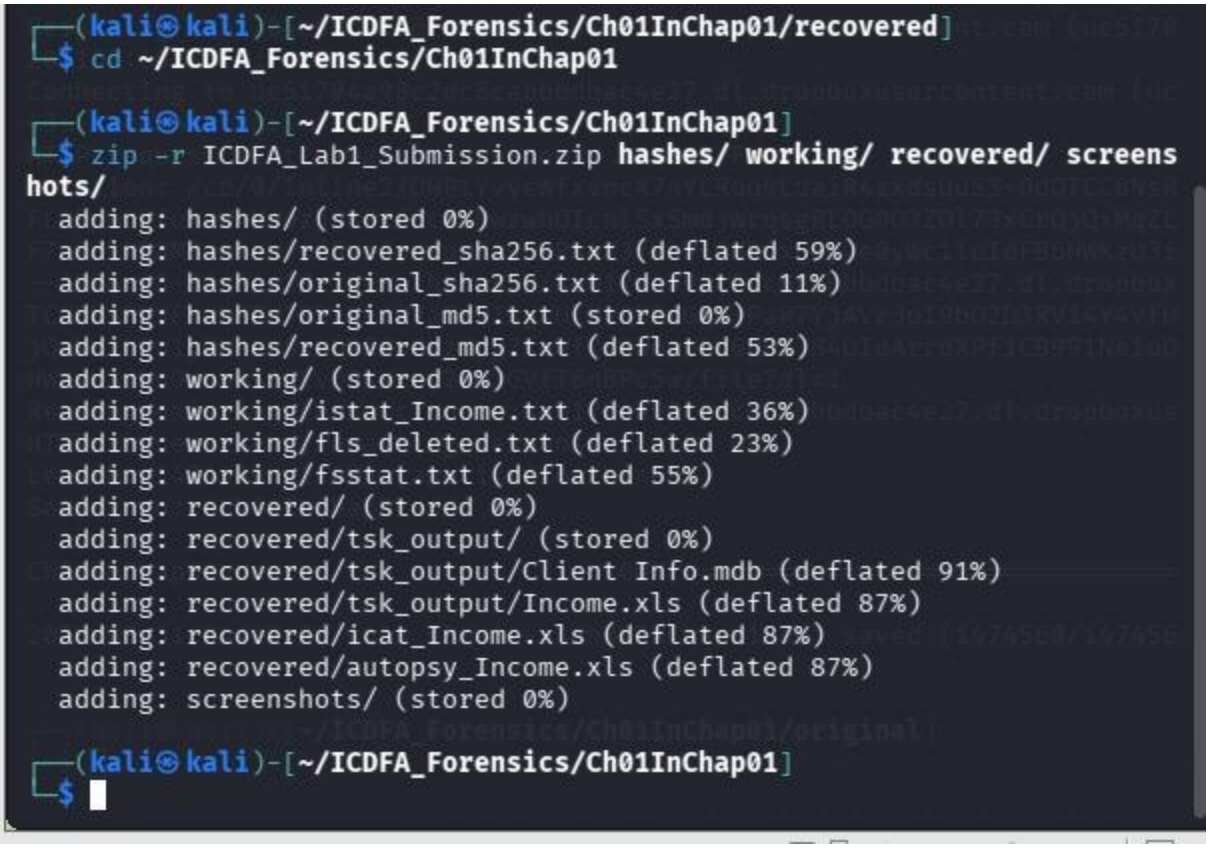


Fig 5.2.5 Automated Recovery (tsk_recover): Executed tsk_recover -a ../original/Ch01InChap01.dd ../recovered/tsk_output/.



Phase 3: Cryptographic Verification & Hash Comparison

To guarantee complete non-repudiation, MD5 and SHA-256 cryptographic hashes were computed across all recovered instances of Income.xls:

Recovery Vector	File Name	MD5 Hash Value	Status
Autopsy GUI Export	autopsy_Income.xls	Matches icat & tsk_recover	**MATCHED**
TSK icat Inode 13	icat_Income.xls	Matches autopsy & tsk_recover	**MATCHED**
TSK tsk_recover	Income.xls	Matches autopsy & icat	**MATCHED**

6. Conclusion

The digital forensics examination of Ch01InChap01.dd was successfully executed. The target deleted file Income.xls was identified at inode 13, extracted via multiple tools, and validated as an uncorrupted Microsoft Excel spreadsheet authored by Amelia Phillips. All submission deliverables, including execution logs, recovered files, and cryptographic reports, were successfully compiled into ICDFA_Lab1_Submission.zip.

7. References

Carrier, B. (2005). File System Forensic Analysis. Addison-Wesley Professional.

The Sleuth Kit Documentation. TSK Command Line Tools (fsstat, fls, istat, icat). https://www.sleuthkit.org/sleuthkit/docs.php

Autopsy Forensic Browser Documentation. Case & Host Management Guide. https://www.sleuthkit.org/autopsy/docs.php